

Copyright 2006 Dr. Brock Williams and Texas Tech University. Unauthorized reproduction prohibited.

1. If the ammount of mayhem is constant in an episode of the Muppet Show, there is a simple formula that gives the amount of mess created:

$$\text{Mess} = \text{Mayhem} * \text{Time}$$

That is, the mess is the product of the mayhem and the length of time it lasts. Typically, however, the mayhem varies over time. In a certain 30 minute episode the amount of mayhem was given by the function $M(t) = 4 \ln(t + 1)$, where $0 \leq t \leq 30$. Describe in complete detail how integration can be used to find the total mess in the episode.

2. It is a well-known fact that the anger generated by a fan during a football game is caused by both the frustration they experience and the length of time it occurs. The anger produced by a constant ammount of frustration is given by the formula

$$\text{Anger} = \frac{1}{3} (\text{Frustration})^2 * \text{Time}.$$

In the first quarter of a game, the Frustration was given by $F(t) = \sin\left(\frac{\pi}{30}t\right)$, where $0 \leq t \leq 15$.

Describe in complete detail how integration can be used to find the total Anger in the first quarter. Discuss the role played by Riemann sums, and set up an integral to find this Anger.

3. A rectangular gate built into a dam is 4 ft high and 8 ft wide. The top of the gate is 2 ft below the surface of the water. What is the force on the gate?
4. Find the mass, moments, and center of mass of a the triangular lamina bounded by the x - and y - axes and $y = 4 - 2x$.